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AERO60492 – Autonomous Mobile Robots

Lecture 1 – Introduction to autonomous mobile robots

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Aims of this unit

- Understand the types of autonomous mobile robots
- Model their motion mathematically
- Learn and implement motion control methods
  - Data stream decoding
  - Feedback control for stability
  - Path planning for guidance
- Understand the sensing, communication and actuation loops
- Legal considerations for aerial robotics

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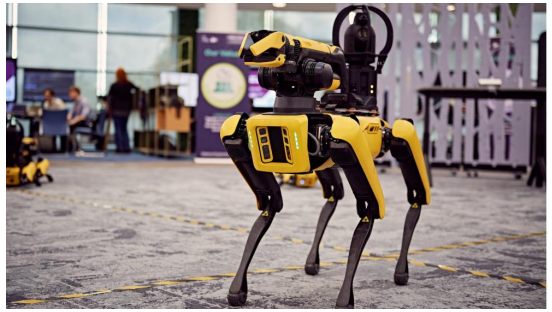
Why?



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Autonomous	Mobile	Robot
“denoting or performed by a device capable of operating without direct human control”	“able to move or be moved freely or easily”	“a machine capable of carrying out a complex series of actions automatically, especially one programmable by a computer”

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Why AMR? 4Ds

Dull

- Routine
- Lack complexity
- Repetitive

Dirty

- Unclean
- Unpleasant
- Contaminated

Dangerous

- High risk
- Harmful
- Hazardous
- Unsafe

Dear

- Costly
- Skilled tasks
- High precision

Types of robots considered

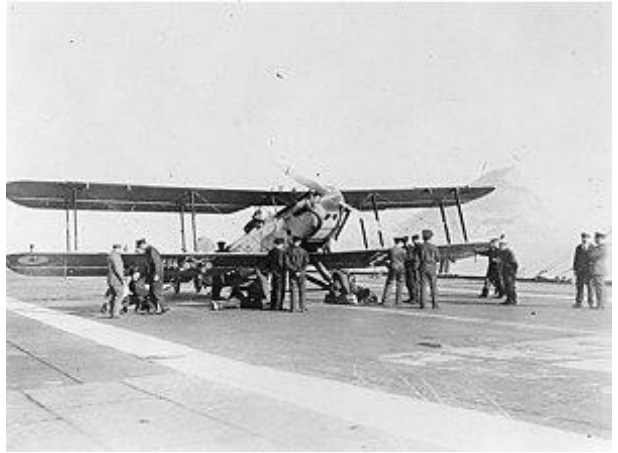
- Anything with locomotion ability
  - Aerial – fly – plane, helicopter, blimp, ornithopter …
  - Aquatic – swim – boat, sail, submarine, …
  - Land – drive – wheeled, tracked, …
  - Land – walk – legs, crawl, …

- Strong focus on Aerial robotics (‘drones’)
- Not covering robotic arms, or anything humanoid. Walkers are on the periphery of the unit and can have many of the methods discussed applied.

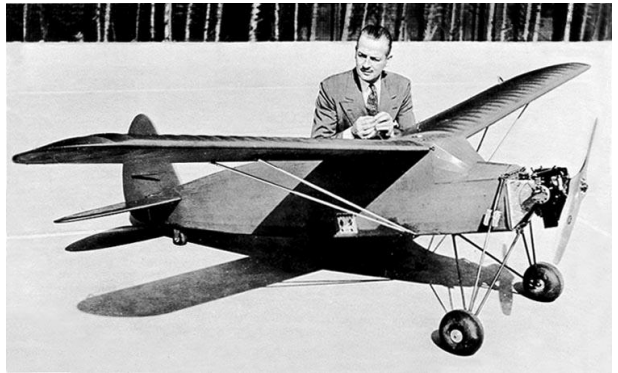
History for UAVs

- 1903 – Wright Brothers first powered human flight
- 1914 – Elmer Sperry develops automatic control using gyroscopic stabilisation for orientation
- 1922 – Lawrence Sperry introduces remote control for setting headings
- 1935 – Reginal Denny develops first dedicated UAV or ‘drone’

The Wright Flyer



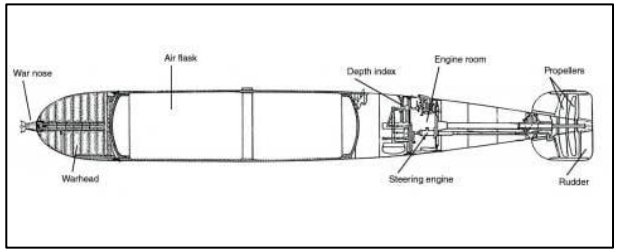
Fairey IIIF



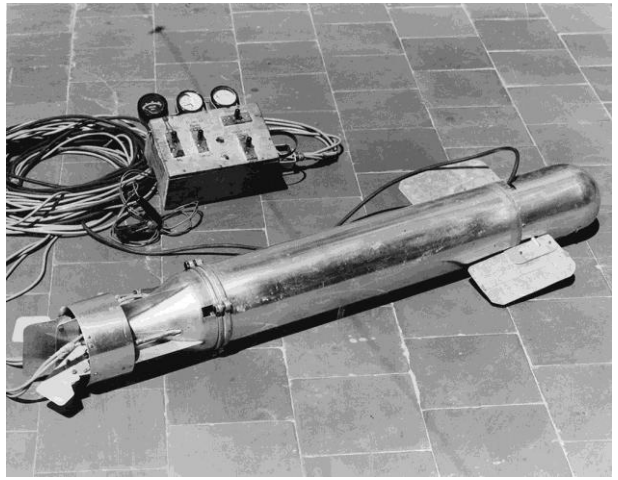
“target drone Denny”

History for UUVs

- 1864 – Luppis-Whitehead Automobile company develops a torpedo using pendulum-and-hydrostat control for stabilisation
- 1954 – Dimitri Rebikoff creates a remote-controlled tethered scooter known as ‘POODLE’
- 1957 – University of Washington deploys the Self-Propelled Underwater Research Vehicle (SPURV)



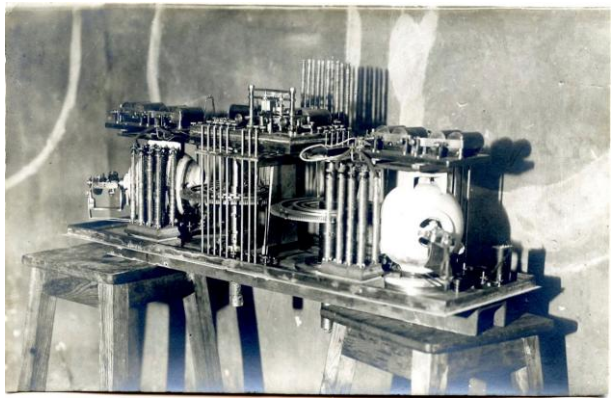
Whitehead torpedo


POODLE

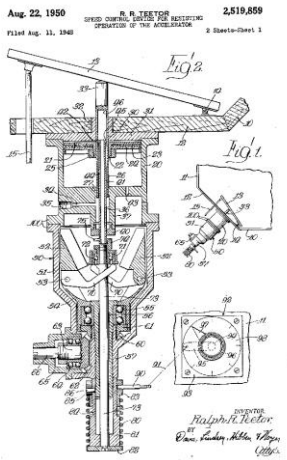
SPURV

History for UGVs

- 1902 – Torres Quevedo creates a remote-control tricycle and boat
- 1948 – Ralph Teetor implements the “Speedostat” which is car cruise control
- 1966 – Stanford Research Institute create ‘Shakey’ the robot for research in automation including early AI



‘Telekino’



Patent for the Speedostat

Shakey with Charles Rosen

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Level of autonomy

The most significant task the system can undertake without constant input from a human operator

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Driverless Cars

https://www.youtube.com/watch?v=BT93bRX2t68

Perception: Reasoning: Planning: Action: Reflection

https://cloud.google.com/discover/what-is-agentic-ai

Levels of autonomy

- None – just a machine

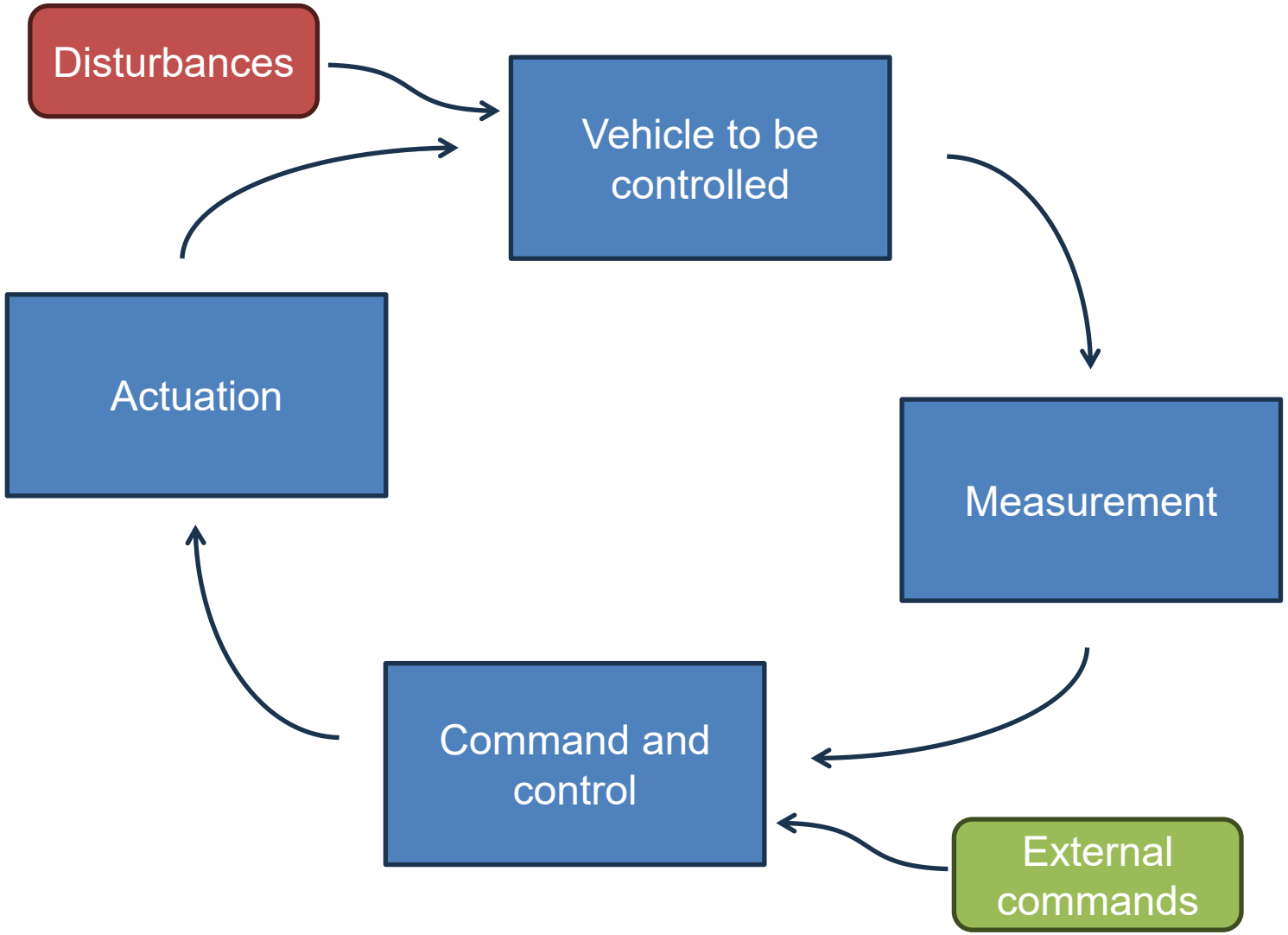
Rigid structure and deterministic

- Automatic – open-loop, scheduled
- Automatic – closed switched, bang-bang
- Automatic – dynamic stabilised, closed-loop

Flexible outcomes and algorithmic

- Autonomous – algorithms, planning, human-in-the-loop
- Autonomous – decision making, human-on-the-loop (supervisory)
- Autonomous – mission oriented, independent of humans.
- Highest – self learning, self-improving

What is automation?



Deterministic vs Non-deterministic

Deterministic

- Fully defined operations for design and execution
- Time-invariant
- Identical output for identical inputs
- Useful for lower levels of autonomy (regulation or stabilisation)

Non-deterministic

- Probabilistic/heuristic
- Stopping criteria
- Not fully defined design
- Higher levels of autonomy (path planning and decision making)

Continuous vs Discrete

Continuous Operation Robots

- Subject to continuous disturbances
- Require constant actuation
- Airborne, water
- 3D

Discrete Operation Robots

- Stop and think
- Disturbances might not change state
- Land
- 2D

Camera drone (used for media filming)

Mobile?	Yes, 3D motion
Continuous/discrete?	Continuous
Level of autonomy?	Mid (stabilisation, obstacle avoidance, commanded)
Dull/Dirty/Dangerous/Dear?	Dear (possibly dangerous)

Rail inspection robot

Mobile?	Yes 1D
Continuous/discrete?	Discrete
Level of autonomy?	Low (automatic motion, pre-programmed)
Dull/Dirty/Dangerous/Dear?	Dull

Bomb disposal crawler

Mobile?	Yes, 2D
Continuous/discrete?	Discrete
Level of autonomy?	Very low (remote control)
Dull/Dirty/Dangerous/Dear?	Dangerous

‘Spot’ robot dog (used for automated search)

Mobile?	Yes, 2D
Continuous/discrete?	Discrete
Level of autonomy?	High (path planning, obstacle avoidance)
Dull/Dirty/Dangerous/Dear?	Dangerous

General Grievous

Mobile?	Yes, 3D
Continuous/discrete?	Discrete
Level of autonomy?	Very high autonomy (decision making and learning)
Dull/Dirty/Dangerous/Dear?	?

Coursework

- Coursework 1 – Serial message decoding 10%
  - Individual work outside labs, individual submission
  - Python script upload
- Coursework 2 – Path planning 15%
  - Individual work with support labs, individual submission
  - Python script upload
- Coursework 3 – Feedback control 25%
  - Group work with drone flight labs, individual submission
  - Video submission

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Coursework Timetable Semester 2

Coursework #	Title	Date set	Date Due
1	Decoding	Week 1: 2nd Feb (Monday)	Week 3: 20th Feb (Friday, 3 pm)
2	Path planning	Week 4: 23rd Feb (Monday)	Week 6: 13th March (Friday, 3pm)
3	Feedback control	Week 7: 16th March (Monday)	Week 12: 15th May (Friday, 3pm)

1. Serial decoding is coursework using a Python environment to decode a binary data file representing a serial data stream. It uses standard Python so can be completed on cluster PCs or student personal computers.
2. Path planning is coursework using a Python environment to implement an A* algorithm. It will use standard Python with some UI libraries. It uses standard Python so can be completed on cluster PCs or student personal computers.
3. Feedback control is a coursework with weekly lab sessions to help develop a motion control for a drone. The submission is a video presentation made independently outside of scheduled sessions. It will make use of a python environment with some more advanced installations and libraries required.

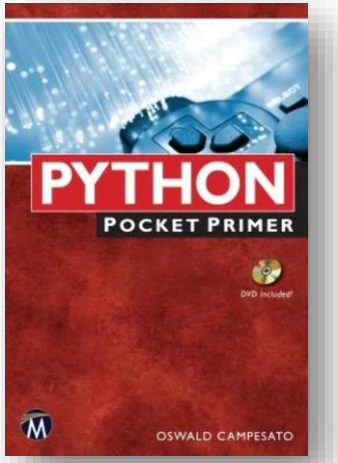
Coursework

- All Python3 based
  - Scripted language
  - Many different distributions
  - Recommend Anaconda (provides basic python, an interface, and popular libraries) https://www.anaconda.com/download
- Familiarise yourself
  - ‘Executing’/running your own script
  - with manipulating basic variables – constants and arrays
  - flow control – `if`, `elif`, `for`, `while`,....

Python Resources

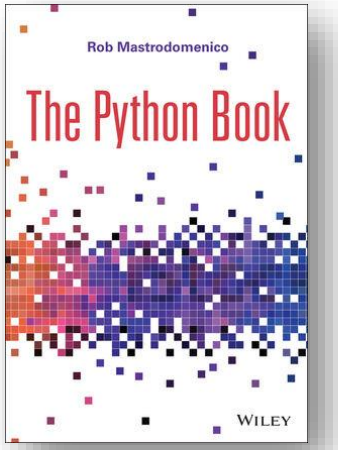
Python have a well written tutorial, and many of the following examples and definitions are abbreviated from this guide:

- https://docs.python.org/3/tutorial/



There are many textbooks available for learning from alternative formats, some examples:

- https://www.librarysearch.manchester.ac.uk/permalink/44MAN_INST/bofker/alma9929861294973016 31
- https://www.librarysearch.manchester.ac.uk/permalink/44MAN_INST/bofker/alma9929841868277016 31



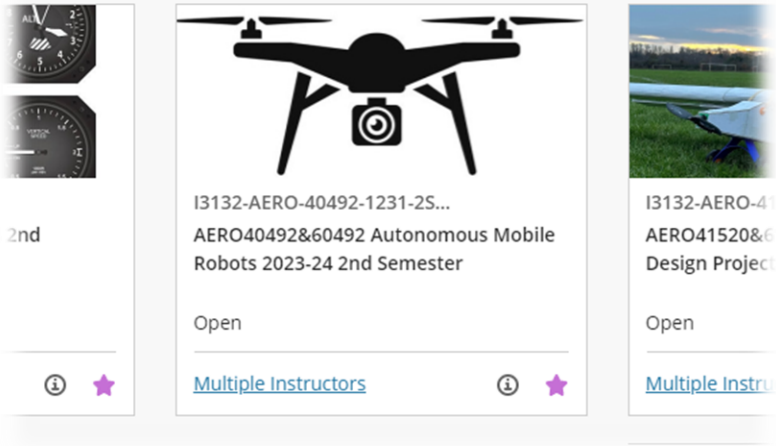
Online videos

- There are many options, but look for ones focusing on the beginner topics covered in these slides.

Coursework

- Most engineering benefits from coding skills
- Robotics its **essential** to have coding skills
- Some marks will be assigned to good coding practices:
  - Clear commenting
  - Efficient use of memory (variables) and CPU (loops etc)
  - Consistent naming – no particular convention needed, but there is a guide https://peps.python.org/pep-0008/
  - Debugging outputs

Content Systems

- Canvas – Lectures, coursework, submissions, announcements, discussions, examples.

- Teams – MSc robotics only for discussions relating to that programme, not this unit.